Extremal Seidel Quadratic Forms on the Boolean Cube

Bounds, Exact Values, Barriers, and Open Problems

Marcus Miccelli

2 August 2026

Abstract

For $n \geq 2$, define

$$F(n) = \min_{a_{ij} \in \{-1,1\}} \max_{x \in \{-1,1\}^n} \left| \sum_{1 \leq i < j \leq n} a_{ij} x_i x_j \right|, \qquad L_n = F(n) n^{-3/2}.$$

The existence and value of $\lim_n L_n$ remain open. This report records a set of internally audited bounds, structural identities, exact computations, counterexamples, and continuation problems. The main universal result is

$$\frac{1}{\pi} \leq \liminf_{n \to \infty} L_n \leq \limsup_{n \to \infty} L_n \leq \frac{1}{2}.$$

The lower endpoint follows from a two-orientation squared-Gram rounding and the upper endpoint from Paley conference matrices and principal restriction. Exact values are reported through order 15. A second part of the report maps the remaining absolute-field problem: it is proved in several diffuse, bounded-component, bounded-degree, and twin-fibre regimes, while a genuinely dense overlapping-correlation regime remains. The report also records barriers that rule out several natural spectral, compositional, superadditive, and low-degree Fourier approaches.

Status. This is an independent research report, not a journal-refereed paper. Statements are labelled as proved, exhaustive computation, numerical, open, or refuted. "Internally audited" means that normalizations and stated dependencies were re-derived or checked independently during the project; it does not replace external peer review. The public repository reproduces the exact search only through order 6. Larger exhaustive results are reported with their historical certificate structure, but the multi-gigabyte search levels are not distributed here.

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1 Problem, normalization, and interpretations

Definition 1.1. A *Seidel matrix* of order n is a matrix $A \in \mathbb{R}^{n\times n}$ such that $A = A^\mathsf{T}$, $A_{ii} = 0$, and $A_{ij} \in \{-1, 1\}$ for $i \neq j$. The set of such matrices is denoted by $\mathcal{S}_n$.

For $A \in \mathcal{S}_n$ and $x \in \{-1, 1\}^n$, put

$$Q_A(x) = \sum_{i<j} A_{ij}x_ix_j = \frac{1}{2}x^\mathsf{T}Ax, \qquad M(A) = \max_x |x^\mathsf{T}Ax|.$$

Also write

$$M_+(A) = \max_x x^\mathsf{T}Ax, \qquad M_-(A) = -\min_x x^\mathsf{T}Ax.$$

The normalization ledger is therefore

$$F(n) = \frac{1}{2}\min_{A\in\mathcal{S}_n} M(A), \qquad L_n = \frac{F(n)}{n^{3/2}}. \tag{1}$$

Every factor of two in this report is with respect to (1).

The problem has several equivalent or neighbouring interpretations. It is a discrepancy problem for a two-colouring of $E(K_n)$, a deterministic Ising ground-state minimization, and the minimum L^∞ norm of a homogeneous order-two Rademacher chaos. Volberg's Boolean Sidon quantity $T_{n,2}$ is exactly $F(n)$ [12]; Astashkin and Lykov establish the $n^{3/2}$ order up to universal constants in a more general weighted setting [1]. The induced-subset discrepancy studied by Erdős and Spencer is related but not identical [4].

Switching $A \mapsto DAD$, where D is diagonal with entries ± 1, and simultaneous row/column permutation both preserve M, $M_\pm$, and the cut quantity below. Thus the natural finite objects are switching classes (two-graphs), modulo permutation when classification is desired.

2 Principal conclusions

Theorem 2.1 (universal window)**.** *For every $n \geq 2$ and every $A \in \mathcal{S}_n$,*

$$M(A) \geq \frac{2}{\pi} n(n-1) \arcsin\left(\frac{1}{\sqrt{n-1}}\right) \geq \frac{2}{\pi} n\sqrt{n-1}. \tag{2}$$

Consequently

$$\liminf_{n\to\infty} L_n \geq \frac{1}{\pi}.$$

There are Seidel matrices of every sufficiently large order, after principal restriction from nearby Paley conference orders, for which

$$F(n) \leq \left(\frac{1}{2} + o(1)\right) n^{3/2}.$$

Hence

$$\frac{1}{\pi} \leq \liminf L_n \leq \limsup L_n \leq \frac{1}{2}.$$

The existence of the limit is not asserted. The exact data do not presently distinguish convergence from persistent arithmetic oscillation.

2.1 Exact values

The following values are classified as EXHAUSTIVE COMPUTATION. Orders 2 through 6 are regenerated by the public code. Orders 7 through 15 come from historical exhaustive or threshold-emptiness computations whose proof structure is summarized in Section 6.

n	2	3	4	5	6	7	8	9	10	11	12	13	14	15
$\min_A M(A)$	2	6	8	8	10	18	20	24	26	34	36	40	42	54
$F(n)$	1	3	4	4	5	9	10	12	13	17	18	20	21	27

The minimizer is unique up to switching and permutation at orders 12, 13, and 14: respectively a doubling of C_6, a vertex deletion of C_{14}, and the Paley conference matrix C_{14}. At order 15, the all-positive one-vertex extension of C_{14} is optimal with $M = 54$. Monotonicity and congruence imply $\min_A M(A) \geq 56$ at order 16.

For the cut discrepancy introduced below, the exact minimum for $n = 3, \ldots, 14$ is

$$2, 4, 4, 5, 8, 10, 12, 13, 16, 18, 20, 21.$$

At order 15 it is either 24 or 26; the exhaustive route needed to separate the two exceeded the available memory.

3 Core identities and lower bounds

Lemma 3.1 (fill-in)**.** *For $A \in \mathcal{S}_n$ and $u \in \{0, \pm 1\}^n$, $|u^\mathsf{T} A u| \leq M(A)$.*

Proof. Complete the zero coordinates of u by independent Rademacher signs to obtain $x \in \{\pm 1\}^n$. All mixed and off-support terms have expectation zero; no diagonal term survives because $A_{ii} = 0$. Thus $u^\mathsf{T} A u = \mathbb{E}[x^\mathsf{T} A x]$, whose absolute value is at most $M(A)$. □

The zero diagonal is essential. For $\left(\begin{smallmatrix} 4 & -1 \\ -1 & -3 \end{smallmatrix}\right)$, the Boolean maximum is 3, but the vector $(1, 0)$ has quadratic value 4.

For a partition $[n] = S \sqcup T$, define

$$\mathrm{Cut}(A) = \max_{S \sqcup T = [n]} \max_{u \in \{\pm 1\}^S,\, v \in \{\pm 1\}^T} u^\mathsf{T} A_{S,T} v.$$

Theorem 3.2 (cut identity)**.** *For every* $A \in \mathcal{S}_n$*,*

$$M_+(A) + M_-(A) = 4\,\mathrm{Cut}(A).$$

In particular, $\max_x |Q_A(x)| \geq \mathrm{Cut}(A)$*.*

Proof. For independently chosen $x, y \in \{\pm 1\}^n$, put $u = (x+y)/2$ and $w = (x-y)/2$. Their supports partition $[n]$ and

$$x^\mathsf{T} A x - y^\mathsf{T} A y = 4 u^\mathsf{T} A w.$$

Conversely every signed cut arises from such a pair. Maximizing x and y independently gives the identity. $\square$

Proposition 3.3 (fixed-split first moment)**.** *For every* $1 \leq s < n$*,*

$$\mathrm{Cut}(A) \geq (n-s)\mathbb{E}|S_s|,$$

where S_s is a sum of s independent Rademacher variables.

Proof. Fix $|S| = s$. For $u \in \{\pm 1\}^S$, optimize the signs on T to obtain $\|A_{S,T}^\mathsf{T} u\|_1$, then average over u. Each of the $n-s$ coordinates is distributed as S_s. $\square$

For odd $s = 2k+1$,

$$\mathbb{E}|S_s| = (2k+1)4^{-k}\binom{2k}{k} \geq \sqrt{\frac{2}{\pi}}\sqrt{s}.$$

The parity restriction matters: the displayed inequality is false for even s. A sharpened Wallis inequality due to Kazarinoff [8], together with a six-residue finite check, absorbs the rounding loss in choosing an odd integer near $n/3$. This gives the finite theorem

$$F(n) \geq \max_{1 \leq s < n} (n-s)\mathbb{E}|S_s| \geq \frac{2\sqrt{2}}{3\sqrt{3\pi}} n^{3/2} = 0.3071059106\ldots\, n^{3/2}. \tag{3}$$

The stronger asymptotic bound is Theorem 2.1.

3.1 Two-orientation squared-Gram rounding

Put $q = n-1$ and $B = A/\sqrt{q}$. Since $B_{ii} = 0$ and $(B^2)_{ii} = 1$, the matrices

$$X_\sigma = \frac{1}{2}(I + \sigma B)^2, \qquad \sigma \in \{-1, 1\},$$

are positive semidefinite correlation matrices. Hyperplane-round each X_σ. For $i \neq j$, write $b = B_{ij}$ and $c = (B^2)_{ij}/2$. The sign-correlation identity yields terms involving $\arcsin(c + \sigma b)$. Averaging the two orientations uses

$$b\big[\arcsin(c+b) - \arcsin(c-b)\big] \geq 2|b| \arcsin|b|.$$

Indeed, the even density $(1-t^2)^{-1/2}$ has least mass on a fixed-length interval when that interval is centered at zero. Summing the $n(n-1)$ identical values $|b| = q^{-1/2}$ proves (2).

3.2 Spectral-edge rounding

Let $\mu(A) = \min\{\lambda_{\max}(A), |\lambda_{\min}(A)|\}$. Applying hyperplane rounding to $I + sA$ and to $I - sA$ at their positive-semidefinite spectral endpoints gives

$$M(A) \geq \frac{2}{\pi} n(n-1) \arcsin\left(\frac{1}{\mu(A)}\right). \tag{4}$$

There is no sign cancellation because $A_{ij} \arcsin(sA_{ij}) = \arcsin s$. The bound is exact for $A = J - I$.

4 Upper bound and conference matrices

For $q \equiv 1 \pmod 4$ a prime power, the symmetric Paley conference matrix C of order $q+1$ satisfies

$$C^2 = qI, \qquad \|C\|_2 = \sqrt{q}, \qquad M(C) \le (q+1)\sqrt{q}.$$

Primes congruent to 1 modulo 4 have relative gaps $o(x)$; the fixed modulus statement follows, for example, from Thorner–Zaman [11]. Restricting a nearby Paley matrix to a principal $n \times n$ block and applying Lemma 3.1 proves the upper half of Theorem 2.1.

Conference matrices are spectrally flat, but their Boolean maxima are not uniformly close to the hyperplane-rounding value. An internally audited local-field calculation gives the following stronger statement.

Theorem 4.1 (conference local improvement)**.** *For every sequence of symmetric conference matrices,*

$$\liminf_n \frac{M(C_n)}{n\sqrt{n-1}} \ge \gamma_{\mathrm{loc}} = 0.6894707435\ldots > \frac{2}{\pi}.$$

If A_n differs from C_n in $h_n = o(n^{3/2})$ unordered edges, the same coefficient holds for A_n.

The stability assertion follows from the exact Lipschitz estimate

$$|M(A) - M(C)| \le 4h.$$

For the local improvement, puncture the normalized conference matrix in the all-ones direction, hyperplane-round, and use the limiting aligned-local-field law

$$L = \sqrt{\frac{2}{\pi}}|G_0| + \sqrt{1 - \frac{2}{\pi}}\, G_1.$$

Independently flipping a negatively aligned site with probability $\min((-L)_+/2, 1)$ gives the stated gain. The traffic input is of the type developed in [10]; the conclusion above is nonperturbative at fixed algorithmic depth.

There is also a coefficientwise conference free-energy theorem. If $H = Q_C/\sqrt{n-1}$, then for every fixed r,

$$\frac{\kappa_{2r}(H)}{n} \longrightarrow (-1)^{r-1}\frac{(2r)!}{4r}\operatorname{Cat}_{r-1}.$$

These are the Taylor coefficients of the formal orthogonal-model function

$$G(\beta) = \frac{1}{4}\left(\sqrt{1+4\beta^2} - 1 - \log\frac{1+\sqrt{1+4\beta^2}}{2}\right).$$

This is *not* a uniform deterministic free-energy comparison: the Taylor series has radius $1/2$, while an improvement over γ_{loc} would require large β. At degree ten the first class-sensitive invariant appears, but its normalized fixed-order contribution is $O(1/n)$; fixed-order universality survives.

5 Structure and arithmetic

Proposition 5.1 (block control)**.** *For $[n] = S \sqcup T$,*

$$\operatorname{Cut}(A_S) + \operatorname{Cut}(A_T) \le \operatorname{Cut}(A) \le \operatorname{Cut}(A_S) + \operatorname{Cut}(A_T) + \|A_{S,T}\|_{\infty\to 1}.$$

The lower inequality combines two block optimizers and chooses a simultaneous sign flip so that the cross term is nonnegative. The upper inequality decomposes a global optimal signed cut into its internal and cross parts.

Principal restriction and arbitrary one-vertex extension give

$$F(n) \le F(n+1) \le F(n) + n.$$

Thus $|L_{n+1} - L_n| = O(n^{-1/2})$, and the set of subsequential limits of (L_n) is a closed interval. These facts alone do not imply convergence.

The quadratic form satisfies

$$x^{\mathsf{T}} A x \equiv n(n-1) \pmod 4, \qquad F(n) \equiv \binom{n}{2} \pmod 2.$$

For odd n, every rectangular cut sum contains an even number of terms, so $\mathrm{Cut}(A)$ is even.

Theorem 5.2 (forced unbalancedness)**.** *If $n \equiv 3 \pmod 4$, then*

$$F(n) > \min_{A \in \mathcal{S}_n} \mathrm{Cut}(A),$$

and no M-minimizing matrix is balanced, where balanced means $M_+(A) = M_-(A)$.

This explains all computed failures of $F(n) = \min_A \mathrm{Cut}(A)$ at $n = 3, 7, 11$ and forces another failure at $n = 15$, independently of the enumeration.

6 Exact computation and provenance

Switching allows the first row and column to be normalized to $+1$ off the diagonal. The public iterator then enumerates

$$2^{(n-1)(n-2)/2}$$

labeled switching-normalized matrices. For each matrix, global-sign symmetry reduces the Boolean cube to 2^{n-1} vectors. This direct method reproduces orders through 6 and is intentionally simple.

The historical computations through order 15 used a more efficient exact integer pipeline: Gray-code energy updates, principal-submatrix pruning from Lemma 3.1, globally canonical switching/permutation certificates, chunked extension levels, and threshold emptiness combined with the congruence above. The order-15 value $M = 54$ was certified by two independent empty searches at thresholds 46 and 50, overlapping certificate sets, and direct evaluation of the witness $[C_{14}, \mathbf{1}]$ on the full antipodal half-cube.

Two implementation failures were discovered and bounded during that work. An early 64-bit certificate encoding overflowed once $\binom{k-1}{2} > 64$; affected levels were recomputed using 128-bit certificates. A fixed-size resume buffer could overflow for sufficiently large seed files; every certified run used smaller per-process seeds, and a patched resume reproduced the relevant levels. These incidents are recorded because they materially affect how the exhaustive claims should be assessed.

The public repository does not ship the multi-gigabyte intermediate levels. Accordingly, values above order 6 use the status labels `historical_exhaustive_audit` or `historical_threshold_emptiness_audit` in `results/reported_exact_values.json`. They should not be mistaken for computations reproducible from this compact release alone.

Heuristic search is not a certificate: at order 15, many annealing restarts stalled at $M = 58$ although the exact optimum is 54.

7 The Hermite residual and the remaining absolute-field problem

The squared-Gram proof can be decomposed beyond its first Hermite chaos. Let y_σ be the rounded sign vector for X_σ, write

$$y_\sigma = \sqrt{\frac{2}{\pi}}\, z_\sigma + \eta_\sigma, \qquad R_\sigma = \mathbb{E}[\eta_\sigma \eta_\sigma^\mathsf{T}],$$

and couple the two orientations using one Gaussian vector g. The arcsine Schur expansion gives, for $n \geq 5$,

$$\frac{R_+ + R_-}{2} \succeq \frac{1 - 2/\pi}{2} I.$$

The paired mean local-field square is at least one, with equality only when $B^2 = I$.

Put

$$u = \frac{y_+ + y_-}{2}, \qquad v = \frac{y_+ - y_-}{2}.$$

Coordinatewise exactly one of u_i, v_i is nonzero:

$$(u_i, v_i) = \begin{cases} (\operatorname{sign} g_i, 0), & |g_i| > |(Bg)_i|, \\ (0, \operatorname{sign}(Bg)_i), & |(Bg)_i| > |g_i|. \end{cases}$$

Define the complementary absolute field

$$U_n = \mathbb{E} \sum_i \max\{|(Bu)_i|, |(Bv)_i|\}.$$

The exact remaining target is

$$\liminf_n \frac{U_n}{n} \geq \sqrt{\frac{2}{\pi}}. \tag{5}$$

A self-normalized cube-flip lemma converts (5) into a universal lower-bound improvement to $0.3233368289\ldots$ for L_n. Separate second-moment floors $\mathbb{E}\|Bu\|_2^2, \mathbb{E}\|Bv\|_2^2 \geq n/2$ do not by themselves imply the required first absolute moment.

Although (5) is open universally, it is proved in a large collection of regimes. Write $D = B^2$ and

$$\overline{\Delta}_n = \frac{1}{n} \sum_{j \neq k} \max\{(n-1)^{-1/2}, |D_{jk}|\}^3.$$

The following statements are classified as PROVEDwithin the project's internal audit.

- (i) If $\overline{\Delta}_n \to 0$, then (5) holds. The same remains true after deleting $o(\sqrt{n})$ exceptional source indices when the complementary cubic mass vanishes.
- (ii) Fixed k-fold twin blow-ups of a globally decorrelated base satisfy the stronger coefficient $\sqrt{2/\pi}\sqrt{(k+2)/3}$. Growing twin fibres are farther from extremality by a direct Boolean lift.
- (iii) Disjoint strong matched pairs, under block-level global decorrelation, satisfy (5).
- (iv) Correlation components of maximum size 265 satisfy (5) under block decorrelation.
- (v) A component partition is unnecessary: a strong-correlation graph of maximum degree 264, with weak off-graph cubic mass, suffices.

(vi) More generally, with

$$\mathcal{L}(\rho) = \frac{2}{\pi}\left(\arcsin\frac{\rho}{2} - \frac{\rho}{2}\right),$$

the maximum-degree condition may be replaced by the oriented-edge density bound

$$\frac{1}{n}\sum_{(j,k)\in E^{\mathrm{or}}} \frac{|\mathcal{L}(D_{jk})|}{|D_{jk}|} \le 3.9698962518\ldots + o(1).$$

The number 265 is a method threshold, not a conjectured phase transition. The unresolved regime must retain total cubic defect mass $\Omega(n)$ and a dense or heavily overlapping strong-correlation graph after every admissible weak-cubic decomposition. Simple concentrated candidates such as twins, matchings, bounded components, and bounded-degree giant components have all been eliminated as obstructions.

8 Barriers and corrected false routes

The following negative conclusions are part of the research output. They prevent a successor from spending effort on arguments whose quantitative ceiling is already understood.

8.1 Spectral and degree-two certificates

Every $A \in \mathcal{S}_n$ has $\|A\|_2 \ge \sqrt{n-1}$. Every diagonal Y satisfying $-Y \preceq A \preceq Y$ has $\operatorname{tr} Y \ge n\sqrt{n-1}$. For the pair of one-sided semidefinite relaxations,

$$\mathrm{SDP}(A)\,\mathrm{SDP}(-A) \ge n^2(n-1),$$

so $\max\{\mathrm{SDP}(A), \mathrm{SDP}(-A)\} \ge n\sqrt{n-1}$, with equality in the maximum floor exactly for conference matrices. Thus spectral, diagonal-scaling, and degree-two SOS certificates cannot certify the desired constant below the spectral ceiling. The symmetric Grothendieck framework gives the relevant norm dictionary [6].

The tempting one-sided claim $\mathrm{SDP}(A) \ge n\sqrt{n-1}$ is false. For the order-5 matrix with one positive off-diagonal edge and all other edges negative, exact primal and dual certificates give

$$\mathrm{SDP}(A) = 9,$$

not the earlier numerical estimate 9.008.

8.2 Composition, nesting, and regularity

Separate PSD-Gram control of bilinear or Kronecker cross terms loses too much to beat the spectral ceiling. The admissible doubling construction also has ratios that increase under iteration. A generic expression $A \otimes B + I \otimes S$ need not even be Seidel: some off-diagonal entries are zero.

For a one-vertex extension $[A, a]$,

$$M([A,a]) = max_y\left(|y^{\mathsf{T}}Ay| + 2|a\cdot y|\right).$$

If the maximizers of A are isotropic, every extension costs at least $2\sqrt{n}$ asymptotically, forcing a nested chain to coefficient at least $4/3$. This applies to several prominent conference-derived seeds.

Switching-regularity alone does not imply spectral attainment. It gives a $\{\pm 1\}$ eigenvector, but the corresponding eigenvalue need not be extremal. The order-9 rook-graph Seidel matrix is regular with exact $M = 24 < 27 = 9\|A\|_2$.

8.3 Single-split and superadditive methods

For a rectangular sign matrix, the fractional relaxation of $\|B\|_{\infty\to 1}$ equals $q\mathbb{E}|S_p|$ exactly. Bowlin's theorem is the classical maximum-frustration form of this statement, including its equality and uniqueness structure [2]. Therefore improvement beyond the first moment is a genuine integrality-gap problem, not an averaging or linear duality problem. The rigorous universal ceiling of the single-split route is $2/(3\sqrt{3}) = 0.384900\ldots$; the often quoted 0.335 is only numerical.

Cut superadditivity and the associated free energy do not imply convergence. An explicit analytic scalar sequence satisfies the relevant monotonicity, increment, window, and superadditivity properties while oscillating. Fekete normalizations fail on exact small values, and bounded-difference concentration has speed n where overcoming the number of sign matrices requires speed n^2.

8.4 Several splits and Fourier summaries

The several-splits functional

$$V(A) = \mathbb{E}_u \max_S \sum_{j\notin S} \left| \sum_{i\in S} A_{ij} u_i \right|$$

lies between the single-split mean and $\mathrm{Cut}(A)$ and is strictly better at several computed orders. It has an exponential L^2 packing, but the desired growing moments do not follow from low-degree Fourier mass.

Several explicit counterexamples delimit this route:

- for every fixed even swap distance h, there are Seidel matrices for which a split-field increment vanishes identically although the field itself is nonconstant;
- dense, subset-robust level-two Fourier mass can coexist with constant modulus, so all L^p norms are equal;
- positive shifted absolute kernels have a level-four multiplier whose sign changes at the natural shift scale, ruling out a common level-four Gram sign;
- low-degree mass, coefficient spread, and restriction robustness alone cannot force $\sqrt{p}$ moment growth.

Any successful continuation must use realizability of the actual shifted absolute-linear-form process or average rigidity over a proportional packed family, not only generic Fourier summaries.

8.5 Other closed shortcuts

Graphon convergence is blind because every candidate at the relevant scale converges to the zero graphon. Guerra-style interpolation has no disorder to integrate and convexification admits the zero matrix. Conference local-field arithmetic gives exact finite ceilings but no uniform asymptotic gap; the compatibility equation $C(Cx) = (n-1)x$ remains essential. Fixed-order conference cumulants agree with the orthogonal model, but that coefficientwise fact does not justify a low-temperature comparison.

9 Related formulations

For the induced-subset discrepancy $D(n)$,

$$D(n) \le F(n) \le 5D(n).$$

Thus the Erdős–Spencer problem has the same order but not the same constant [4].

The coding formulation identifies $F(n)$ with the distance/covering-radius problem for first-order Reed–Muller functions restricted to the Hamming weight-two slice; general weightwise nonlinearity is discussed by Gini and Méaux [7], but no exact $k = 2$ asymptotic constant is known.

For $B \in \{\pm 1\}^{p \times q}$,

$$\min_B \|B\|_{\infty \to 1} = pq - 2F_{\max}(K_{p,q}),$$

where the right side is maximum frustration of a signed complete bipartite graph. Bowlin gives exact formulas for $p \leq 7$ [2]; the proportional regime remains open. This is closely related to the Gale–Berlekamp switching game [5, 3, 9], but that game is bilinear and its constants do not directly transfer to the symmetric quadratic problem.

10 Open problems and continuation map

Open problem 10.1 (limit)**.** Does $\lim_n L_n$ exist? If so, what is its value?

Open problem 10.2 (absolute field)**.** Prove (5) for the remaining dense overlapping correlation regime, or construct a counterexample. A proof would improve the universal lower endpoint beyond $1/\pi$.

Open problem 10.3 (infinite sub-spectral family)**.** Is there a $\delta > 0$ and an infinite sequence $A_n \in \mathcal{S}_n$ such that $M(A_n) \leq (1 - \delta)n^{3/2}$?

Open problem 10.4 (rectangular integrality gap)**.** Determine the proportional asymptotic of $\min_B \|B\|_{\infty \to 1}/(q\sqrt{p})$. The first concrete finite case beyond Bowlin's formulas is $p = 8$, in particular the 8×16 problem.

Open problem 10.5 (order-15 cut)**.** Is $\min_{A \in \mathcal{S}_{15}} \operatorname{Cut}(A)$ equal to 24 or 26? A structural argument is preferable to repeating the infeasible threshold-96 enumeration. A value-24 witness would have all fifteen order-14 deletions highly unbalanced.

Open problem 10.6 (Bernoulli lower-tail rate)**.** Does

$$\lim_{n \to \infty} n^{-2} \log_2 \#\{A \in \mathcal{S}_n : M(A) \leq \lambda n^{3/2}\}$$

exist? Gaussian or typical-scale universality cannot by itself determine this Bernoulli support-endpoint problem.

Open problem 10.7 (growing-order conference comparison)**.** Can the fixed-cumulant conference universality be upgraded to a deterministic comparison uniform in a growing order or inverse temperature? Any such argument must control the class-sensitive irreducible cores rather than discard them termwise.

11 Reproduction guide

The compact public computation is intentionally modest:

```
python -m pip install -e .
python -m unittest discover -s tests -v
python experiments/exact_small_n.py --max-order 6 \
  --output results/exact_small_n.json
```

It validates Seidel inputs, evaluates $M(A)$ on the antipodal half-cube, enumerates every first-row-normalized labeled switching representative, and computes $\min_A M(A)$ exactly through order 6.

The compact report source and the exact-result JSON are intended to be stable handoff points. Large future computations should publish a certificate format, source revision, deterministic regeneration command, hashes, and an independent evaluator before publishing a claimed optimum.

12 Final status

The limit problem is unresolved. The strongest universal information is the window $[1/\pi, 1/2]$, exact values through order 15, and a sharply narrowed description of what a counterexample to the remaining absolute-field target must look like. Conference matrices and their sparse perturbations are not extremal obstructions; neither are diffuse correlations, twins, matchings, bounded correlation components, or bounded-degree strong-correlation graphs. The live problem is concentrated in dense overlapping correlation structure or in a new construction principle not captured by the audited barriers.

References

[1] Sergey V. Astashkin and Konstantin V. Lykov. Random unconditional convergence of Rademacher chaos in L_∞ and sharp estimates for discrepancy of weighted graphs and hypergraphs. *Mathematische Annalen*, 2025.

[2] Garry S. Bowlin. Maximum frustration in bipartite signed graphs. *The Electronic Journal of Combinatorics*, 19(4):P10, 2012.

[3] Jeffrey Carlson and Daniel Stolarski. The correct solution to Berlekamp's switching game. *Discrete Mathematics*, 287:145–150, 2004.

[4] Paul Erdős and Joel Spencer. Imbalances in k-colorations. *Networks*, 1(4):379–385, 1971.

[5] Peter C. Fishburn and Neil J. A. Sloane. The solution to Berlekamp's switching game. *Discrete Mathematics*, 74:263–290, 1989.

[6] Shmuel Friedland and Lek-Heng Lim. Symmetric Grothendieck inequality. 2020.

[7] Aline Gini and Pierrick Méaux. Weightwise nonlinearity of boolean functions. *Discrete Applied Mathematics*, 322:320–341, 2022.

[8] Donat K. Kazarinoff. On Wallis' formula. *Edinburgh Mathematical Notes*, 40:19–21, 1956.

[9] Daniel Pellegrino and Daniel Jr. Raposo. Upper bounds for the constants of Bennett's inequality and the Gale–Berlekamp switching game. *Mathematika*, 2024.

[10] Lucas Pesenti. Universality of first-order methods on random and deterministic matrices. Manuscript, `https://lucaspesenti.github.io/universality.pdf`, 2026.

[11] Jesse Thorner and Asif Zaman. Refinements to the prime number theorem for arithmetic progressions. *Mathematische Zeitschrift*, 306(3):54, 2024.

[12] Alexander Volberg. An estimate of Sidon constant for complex polynomials with unimodular coefficients. 2022.